Q1

(a) Here,

\# nodes = 2

$\therefore$ degree = nodes -1

$= 2-1$

$= 1$

$\therefore$ For getting a 3 degree polynomial, we need to use Hermite Interpolation. where Hermite degree = $2n+1$ where n = normal degree. Thus,

Hermite degree = $2n+1$

$= 2\cdot 1+1 = 2+1$

$= \underline{\underline{3}}$

(b) given, $f(x) = x\ln x \Rightarrow f'(x) = x\cdot\frac{1}{x} + \ln x = 1+\ln x$

x	$f(x)$	$f'(x)$
$x_0=1$	0	1
$x_1=5$	8.0472	2.6095

$$\therefore P_3(x) = h_0(x)\cdot \cancel{f(x_0)}^{\,0} + h_1(x) f(x_1) + \hat{h}_0(x)\cancel{f'(x_0)}^{\,1} + \hat{h}_1(x) f'(x_1)$$

$$= h_1(x) f(x_1) + \hat{h}_0(x) + \hat{h}_1(x) f'(x_1)$$

$$\therefore h_1(x) = \{1-2(x-x_1)\, l_1'(x)\}\{l_1(x)\}^2$$

$$= \{1-2(x-5)\tfrac{1}{4}\}\{\tfrac{x}{4}-\tfrac{1}{4}\}^2$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0}$$

$$= \frac{x-1}{5-1}$$

$$= \frac{x-1}{4} = \frac{x}{4}-\frac{1}{4}$$

$$l_1'(x) = \frac{1}{4}$$

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$\hat{h}_0(x) = (x - x_0)\{l_0(x)\}^2$

$= (x-1)\{-\frac{x}{4} + \frac{5}{4}\}^2$

$l_0(x) = \frac{x - x_1}{x_0 - x_1}$

$= \frac{x-5}{1-5}$

$= \frac{x-5}{-4}$

$= -\frac{x}{4} + \frac{5}{4}$

$\hat{h}_1(x) = (x - x_1)\{l_1(x)\}^2$

$= (x-5)\{\frac{x}{4} - \frac{1}{4}\}^2$

$\therefore P_3(x) = \{1 - 2(x-5)\frac{1}{4}\}\{\frac{x}{4} - \frac{1}{4}\}^2 \times 8.0472 + (x-1)\{-\frac{x}{4} + \frac{5}{4}\}^2 +$

$(x-5)\{\frac{x}{4} - \frac{1}{4}\}^2 \times 2.6095$ Ans

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Q2

Given, $f(x) = \frac{1}{1+25x^2}$

interval $[-4, 6]$, radius $= 5 = A$, center $= 1 = c$

degree $= 2 \rightarrow$ nodes $= 2+1 = 3$

$\therefore$ (a) Equal angles: $\theta_j = \frac{(2j+1)\pi}{2(n+1)}$

$\therefore \theta_0 = \frac{(2\cdot 0+1)\pi}{2(2+1)} = \frac{1}{6}\pi$

$\therefore \theta_1 = \frac{1}{2}\pi$

$\therefore \theta_2 = \frac{5}{6}\pi$

(b) Chebyshev Node, $x_j = A\cos(\theta_j) + c$

$\Rightarrow x_0 = 5\cos(\theta_0) + 1$

$= 5.330127019$

(Ans)

$\Rightarrow x_1 = 5\cos(\theta_1) + 1$

$= 1$

$\Rightarrow x_2 = 5\cos(\theta_2) + 1$

$= -3.330127019$

Q3

Here,
Step size $= 0.1$

$x_0 + h = 1.2 + 0.1 = 1.3$

$x_0 - h = 1.2 - 0.1 = 1.1$

Central Difference, $CD = \dfrac{f(x_0+h) - f(x_0-h)}{2h}$

$$\Rightarrow f'(1.2) = \frac{f(1.3) - f(1.1)}{2 \times 0.1}$$

$$= \frac{0.01131 - 0.2902}{2 \times 0.1}$$

$$= -1.39445 = -1.394 \text{ (Ans)}$$

Backward Difference $= \dfrac{f(x_0) - f(x_0-h)}{h}$

$$\Rightarrow f'(1.2) = \frac{f(1.2) - f(1.1)}{0.1}$$

$$= \frac{0.1669 - 0.2902}{0.1}$$

$$= -1.233 \text{ (Ans)}$$

Given, $f(x) = x\cos(x) - x + \sin x$

$$\therefore f'(x) = x \cdot -\sin(x) + \cos x - 1 + \cos x$$

$$= 2\cos x - x\sin x - 1$$

(with)

$$\therefore f'(1.2) = -1.393731394$$

$$= -1.393\ldots = -1.394$$

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$\therefore$ Relative Error, $RE_{CD} = \left| \frac{-1.394 + 1.394}{-1.394} \right| \times 100\%$

$= 0\%$ (Ans)

$\therefore$ Relative Error, $RE_{BD} = \left| \frac{-1.394 + 1.233}{-1.394} \right| \times 100\%$

$= 11.53\%$ (Ans)

$f(x) = x\cos x - x + \sin x$

$f'(x) = 2\cos x - x\sin x - 1$ | $x \in [1.1 - 1.3]$

$f''(x) = -2\sin x - x\cos x - \sin x$

$= -x\cos x - 3\sin x$

$f''(\xi) = |-x\cos x - 3\sin x|$

$= x\cos x + 3\sin x$

$= 1.3\cos(1.1) + 3\sin(1.3)$

$= 3.480$

$\therefore BD = \left|\frac{3.480 \times (-0.1)}{2}\right| = 0.1740$ Ans

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Q1

(i) $D_h = \frac{f(x+h) - f(x-h)}{2h}$

From Taylor series,

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)(x-x_0)^3}{3!} + \cdots$$

Now,

$$f(x_1+h) = f(x_1) + \frac{f'(x_1)(x_1+h-x_1)}{1!} + \frac{f''(x_1)(x_1+h-x_1)^2}{2!} + \frac{f'''(x_1)(x_1+h-x_1)^3}{3!} + \frac{f^{(4)}(x_1)(x_1+h-x_1)^4}{4!} + \cdots$$

$$\Rightarrow f(x_1+h) = f(x_1) + f'(x_1)h + \frac{f''(x_1)h^2}{2!} + \frac{f^{(3)}(x_1)h^3}{3!} + \frac{f^{(4)}(x_1)h^4}{4!} + \frac{f^5(x_1)h^5}{5!} + O(h^6) \quad \text{(I)}$$

$$\Rightarrow f(x_1-h) = f(x_1) - f'(x_1)h + \frac{f^2(x_1)h^2}{2!} - \frac{f^3(x_1)h^3}{3!} + \frac{f^4(x_1)h^4}{4!} - \frac{f^5(x_1)h^5}{5!} + O(h^6) \quad \text{(II)}$$

(I) − (II)

$$2f'(x_1)h + 2\cdot\frac{f^3(x_1)h^3}{3!} + 2\,f^5(x_1)h^5 + O(h^7)$$

$$D_c = \frac{1}{2h}\left(2f'(x_1)h\right) + \cancel{2f'(x_1)h} + 2\frac{f^3(x_1)h^3}{3!} + 2\frac{f^5(x_1)h^5}{5!} + O(h^7)$$

$$\Rightarrow D_h = f'(x_1) + \frac{f^3(x_1)h^2}{3!} + \frac{f^5(x_1)}{5!}h^4 + O(h^6) \quad \text{— (III)}$$

Replacing h with $4h/3$ in (III).

$$D_{\frac{4h}{3}} = f'(x_1) + \frac{f^3(x_1)}{3!}\left(\frac{4h}{3}\right)^2 + \frac{f^5(x_1)}{5!}\left(\frac{4h}{3}\right)^4 + O(h^6) \quad \text{— (IV)}$$

$$\Rightarrow \left(\tfrac{3}{4}\right)^2 \times D_{4h/3} = \left(\tfrac{3}{4}\right)^2 f'(x_1) + \frac{f^3(x_1)}{3!}\left(\frac{4h}{3}\right)^2 + \frac{f^5(x_1)}{5!}\left(4h/3\right)^4 + O(h^6) \quad \text{— (V)}$$

(V − III)

$$D_h - \left(\tfrac{3}{4}\right)^2 \times D_{4/3} = \left(\left(3/4\right)^2 - 1\right) f'(x_1) + \frac{f^5(x_1)h^4}{5!}\left(\left(4/3\right)^2 - 1\right) + O\left(h^6\left(\frac{4^4}{3^4} - 1\right)\right)$$

$$\Rightarrow f'(x_1) = \frac{16/9\ D_{4h/3} - D_h}{3^2/4^2 - 1}$$